ICTD COURSE CASE STUDY

Studying Power Shifts in Community-Centric AI: A Methodology

How to conduct a socio-technical dynamics analysis of a deployed technology — using Commons Connect as the running case

Based on: Ganapathy, A., Heeks, R. & Iazzolino, G. "Materialising Equity in Community-Centric AI Tools: The Case against Technocracy." Centre for Digital Development Working Paper 117, University of Manchester, 2026.

July 2026

A Five-Step Method for Studying Power Dynamics

1	**Map the stakeholder ecosystem**	*Who is embedded in this system, and between whom might power shift?*
2	**Turn the map into research questions**	*Which of these possible shifts will this study actually examine?*
3	**Collect data**	*Interviews, focus group discussions (FGDs), field observation, documents*
4	**Code and analyse thematically**	*Transcripts → codes → themes → connected back to the RQs*
5	**Discuss in a broader context**	*How do these findings speak to other cases and to theory?*

PART 1

The Case, Briefly

Same tool as Decks 1–2, different lens: not whether Commons Connect improves water outcomes, but whether its usage process shifts power more equitably.

Maps to: paper Sections A (Introduction) and C1 (Case description)

Commons Connect, Revisited

- Deck 1 covered Commons Connect as a technical and participatory design case (Sections 3–5 of the ICTD 2024 paper): datasets, indicators, field testing.
- **This paper studies the same tool through a completely different question: not "does the tool produce better water-security outcomes," but "does deploying this tool shift power more equitably among the people embedded in the system around it?"**
- Commons Connect is part of CoRE Stack (Commoning for Resilience and Equality), launched 2023 by CommonsTech Foundation for Participatory Technologies (CFPT), a non-profit based at IIT Delhi.
- At the time of this study, it was being piloted with CSO partners across nine locations in rural India, generating Detailed Project Reports (DPRs) for MGNREGA natural-resource-management planning.
- *This is a second, complementary lens on Topic 4 (Socio-technical Dynamics at Scale) — Deck 1 already touched equity mapping and stakeholder politics (Sections 5.3–5.4); this paper makes that the entire object of study, with a dedicated qualitative methodology.*

A Fundamentally Different Kind of RQ

- **The paper's RQ: "Under what conditions can the design and implementation of AI enable more equitable terms of incorporation for communities that it intends to benefit?"**
- Notice what this is not asking:
  - Not: did water security improve? (that's Deck 2's RCT territory)
  - Not: is the site-suitability model accurate? (that's Deck 1's computing-fundamentals territory)
  - **Instead: as this tool gets used, who gains decision-making power, whose knowledge counts as legitimate, and whose position in the system changes — and under what conditions?**
- *This is a question about relationships and institutions, not about model accuracy or measured impact — and it needs a correspondingly different method: qualitative, stakeholder-centred, and theory-guided rather than experimental.*

PART 2

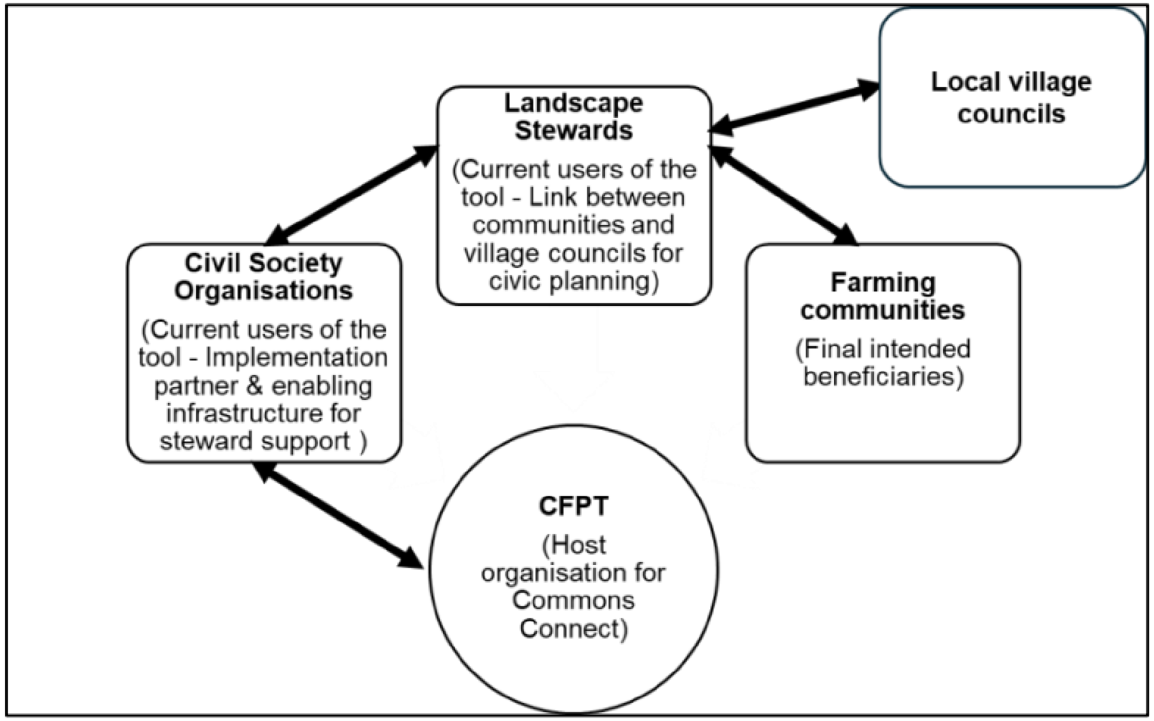
Step 1 — Mapping the Stakeholder Ecosystem

Before you can ask what shifted, you have to know who is actually in the room — and what relationships between them are even capable of shifting.

Maps to: paper Section C1 (Case description) and Figure 3

Who Is Embedded in This System?

- Five identifiable actor groups sit around Commons Connect, each with a distinct role, incentive, and relationship to the tool:
  - CFPT — the host organisation; owns the tool, employs the data scientists who build the geospatial datasets
  - CSOs (Civil Society Organisations) — implementation partners; enabling infrastructure for steward support
  - Landscape stewards — the primary hands-on users; youth/resource persons from the community, link between village councils and communities
  - Farming communities — the final intended beneficiaries
  - Local village councils / Panchayats — the governance bodies that approve and act on plans generated by the tool
- **This is not a list for its own sake — every arrow between these boxes is a place where power could concentrate or redistribute.**



Where Could Power Plausibly Shift?

Relationship	A priori equity concern
CFPT ↔ CSOs / stewards	Who controls the design of datasets and indicators — does technical authority concentrate at the top?
CSOs / CFPT ↔ landscape stewards	Resource and skill asymmetry: stewards depend on CSOs/CFPT for training to meaningfully use the tool
Landscape stewards ↔ communities	Mediation risk: stewards interpret and translate data — whose voice actually reaches the village council?
Communities (internal) ↔ village councils	Institutional legitimacy: does the tool change who is heard in Gram Sabha meetings, or just who is counted?
Communities ↔ MGNREGA / the state	Does a stronger evidence base actually change fund allocation, or just produce better-documented exclusion?

This table is built before data collection — it is the study's a priori map of "places worth looking," not a summary of findings.

Mapping First, Findings Second

- The methodological discipline here is sequencing: the stakeholder map and its a priori equity concerns are drawn up before the interviews happen, grounded in the theoretical framework the study adopts (Heeks' 2022 adverse/advantageous digital incorporation framework — the paper's Section B, which this deck does not cover in depth).
- **Without this step, a qualitative study risks being reverse-engineered: only reporting relationships the researcher happened to notice, with no way to know what was missed.**
- With this step, every research question in the next stage can be traced back to a specific relationship on the map — and the write-up can be honest about which mapped relationships were, or weren't, well covered by the data actually collected.
- *For your own course projects: this is the step to do first, on paper, before writing a single interview question.*

PART 3

Step 2 — From Stakeholder Map to Research Questions

The stakeholder map produces many possible questions. A study has to decide which relationships it will actually examine, and through what conceptual lens.

Maps to: paper Section B (theoretical framework, briefly) and the RQ in Section A

One Broad RQ, Several Study-able Dimensions

- The broad RQ ("under what conditions can AI enable more equitable incorporation?") is too abstract to code data against directly — it needs to be broken into dimensions that map onto specific, observable practices.
- This study uses the ADI+ (advantageous digital incorporation) framework as its organising lens, which supplies four dimensions directly traceable to the stakeholder-map relationships from Step 1:
  - Design equity — who controls how the system is built (CFPT/CSO ↔ stewards/communities)
  - Resource equity — who has the skills/capacity to meaningfully use it (CFPT/CSO ↔ stewards)
  - Institutional equity — whose participation is treated as legitimate (communities ↔ village councils)
  - Non-coercion — whether engaging with the tool is genuinely voluntary (all actors ↔ the tool itself)
- *The framework is a lens for organizing the analysis, not a hypothesis being tested — the study still asks what CFPT/CSO/steward/community practices actually look like, and only afterward interprets them through these four dimensions.*

From Broad RQ to Coded Sub-Questions

Broad RQ (paper, Section A)	Operationalized sub-question actually coded for
Under what conditions can AI enable more equitable incorporation?	Does the design process foreground structural inequities, or lead with technical capability? (D1.1)
	Does the tool's data get contextualised with local/community knowledge, or treated as a complete picture? (D1.2)
	Does using the tool change who is seen as legitimate in village planning meetings? (D1.3)
	Is there deliberate investment in stewards' capacity to use and interpret the tool? (D1.4)
	Is engagement with the tool's outputs voluntary, contestable, and non-binding? (D1.5)

Each row on the right becomes one of the paper's D1.x findings — this is the direct bridge between Step 2 (research questions) and Step 4 (coded themes) later in this deck.

PART 4

Step 3 — Data Collection

Interviews, focus groups, field observation, and documents — triangulated across four participant groups and nine field sites.

Maps to: paper Section C2.1 (Data collection)

Four Data Collection Methods, Triangulated

- Document / website review — CFPT's internal and public-facing materials on Commons Connect, used for triangulation against what informants said.
- In-depth semi-structured interviews — 1:1, 30–50 minutes, conducted in Hindi and English.
- Focus group discussions (FGDs) — 45–60 minutes, some conducted in regional languages with CSO staff facilitating live translation.
- Field observations — at two pilot sites still in early rollout (Dhanpur and Morchand, Gujarat), capturing the tool in active use rather than only recalled in interview.
- **Interviews and FGDs were audio-recorded with consent, transcribed, translated into English where needed, and combined with same-day field notes before being loaded into NVivo for coding.**

58 Informants, Four Participant Groups

58	11	7	9
total key informants	in-person interviews	focus group discussions	field locations covered

Participant group	Number	Role in the ecosystem
CFPT staff	7	Tool owners / data scientists
CSO staff	8	Implementation partners
Landscape stewards	18	Primary hands-on users
Community members	25	Final intended beneficiaries

Notice: community members (25) are the largest group interviewed, but almost entirely through FGDs (47 of 58 informants were FGD-only) rather than 1:1 interviews — a coverage pattern that resurfaces directly in the Limitations discussion later in this deck.

Locations, Languages, and Practicalities

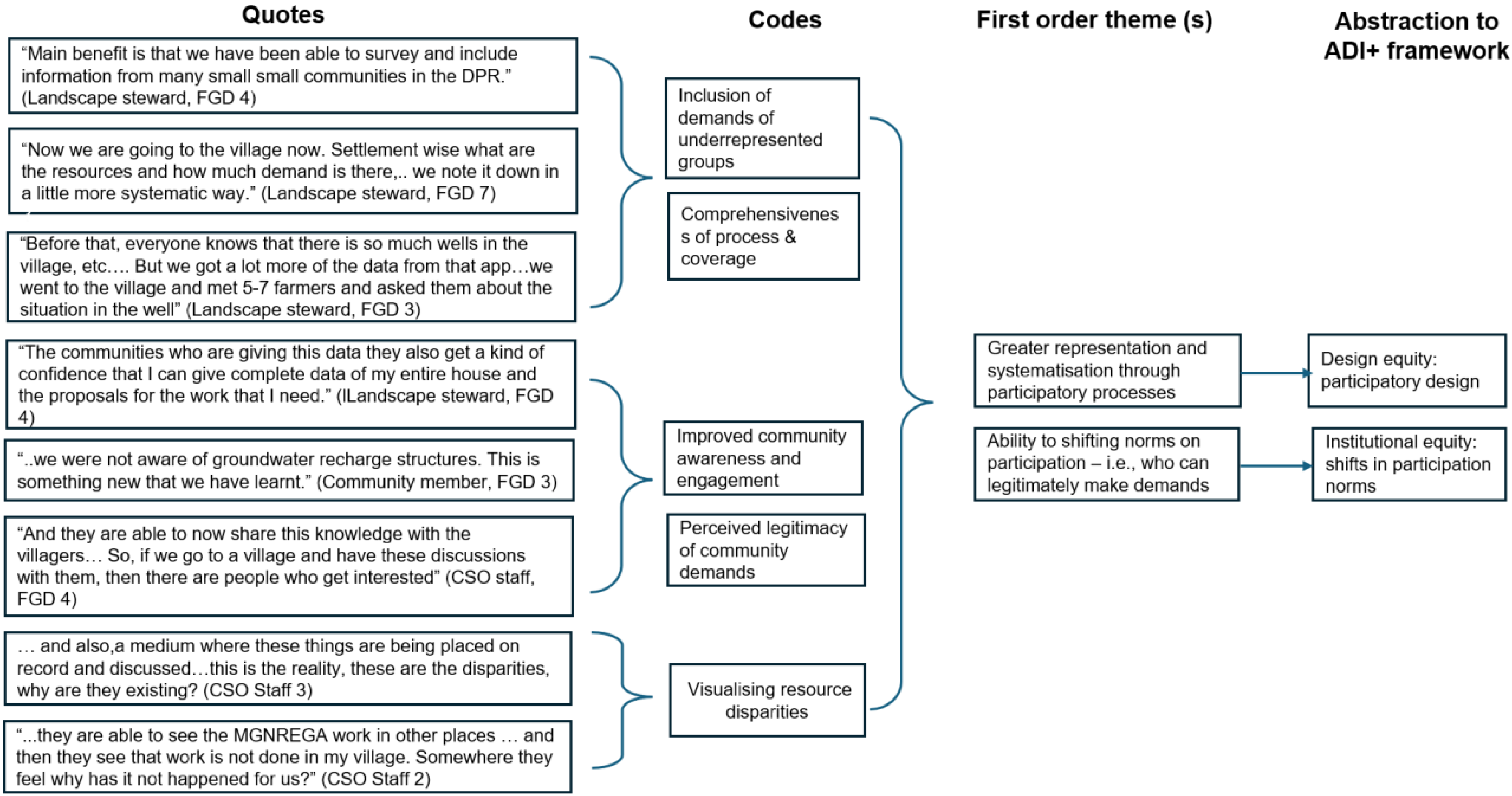
- Nine locations across four states: Gujarat (Dahod, Bhavnagar — early rollout), Jharkhand (Dumka), Rajasthan (Bhilwara), Odisha (Angul — completed pilots).
- **Note the overlap with Deck 1's field-testing sites (Masalia/Dumka, Angul, Pindwara/Rajasthan) — largely the same CSO partnerships (FES, SUPPORT, PRADAN) studied from a different angle a year or so later.**
- In three Angul FGDs, discussions ran in the regional language with CSO staff live-translating to/from Hindi — a real methodological dependency worth naming: the researcher's access to community voice was partly mediated by CSO staff, even at the point of data collection.
- First author was the sole interviewer/facilitator, conversant in Hindi and English; CFPT connected her to CSO partners, who organised the field visits.

PART 5

Step 4 — Coding & Thematic Analysis

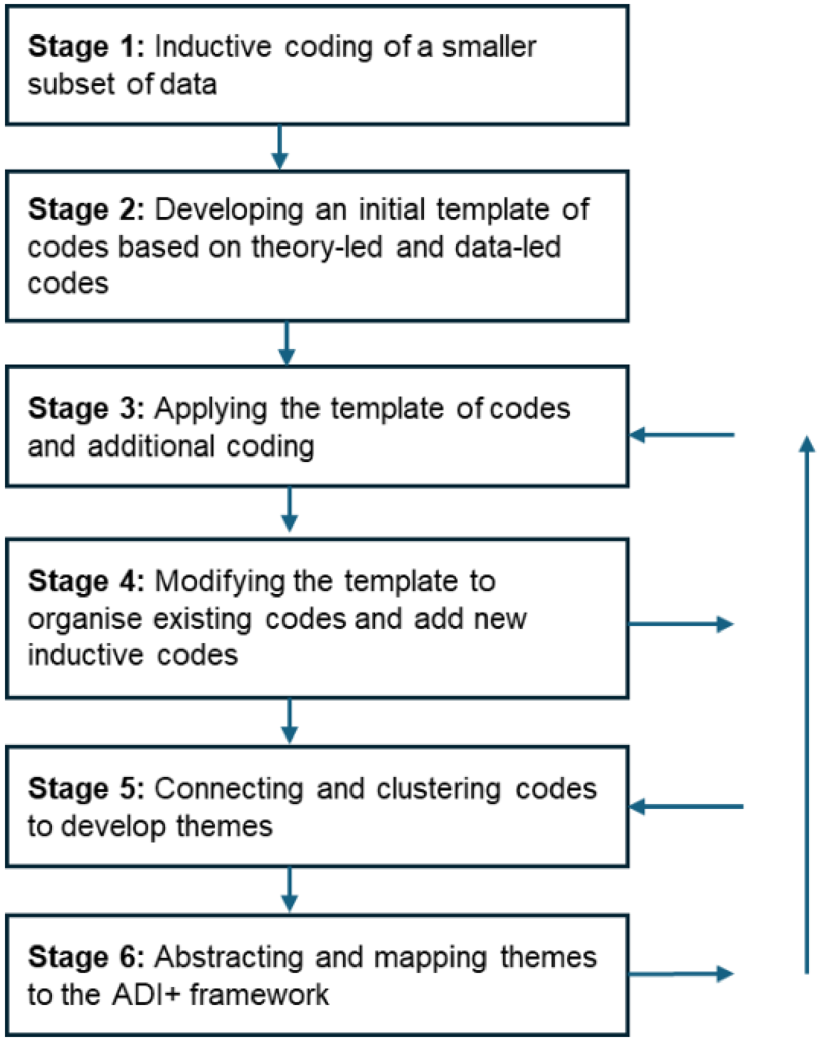
Transcripts become codes, codes become themes, themes get connected back to the research questions — using a disciplined hybrid inductive/deductive method.

Maps to: paper Section C2.2 (Data analysis) and Figures 5–6


Quotes
Codes
First order theme (s)
Abstraction to ADI+ framework
"Main benefit is that we have been able to survey and include information from many small small communities in the DPR." (Landscape steward, FGD 4)
"Now we are going to the village now. Settlement wise what are the resources and how much demand is there,.. we note it down in a little more systematic way." (Landscape steward, FGD 7)
"Before that, everyone knows that there is so much wells in the village, etc…. But we got a lot more of the data from that app…we went to the village and met 5-7 farmers and asked them about the situation in the well" (Landscape steward, FGD 3)
"The communities who are giving this data they also get a kind of confidence that I can give complete data of my entire house and the proposals for the work that I need." (lLandscape steward, FGD 4)
"..we were not aware of groundwater recharge structures. This is something new that we have learnt." (Community member, FGD 3)
"And they are able to now share this knowledge with the villagers… So, if we go to a village and have these discussions with them, then there are people who get interested" (CSO staff, FGD 4)
… and also,a medium where these things are being placed on record and discussed…this is the reality, these are the disparities, why are they existing? (CSO Staff 3)
"...they are able to see the MGNREGA work in other places … and then they see that work is not done in my village. Somewhere they feel why has it not happened for us?" (CSO Staff 2)
Inclusion of demands of underrepresented groups
Comprehensiveness of process & coverage
Improved community awareness and engagement
Perceived legitimacy of community demands
Visualising resource disparities
Greater representation and systematisation through participatory processes
Ability to shifting norms on participation – i.e., who can legitimately make demands
Design equity: participatory design
Institutional equity: shifts in participation norms

A Hybrid Inductive + Deductive Approach

- Two established qualitative traditions are combined:
  - Gioia et al. (2013) — data-driven, inductive: let first-order concepts emerge directly from informants' own words
  - King (2012) — deductive template analysis: use the theoretical framework (ADI+) to build an a priori coding template
- Six stages: (1) inductive coding of a subset of transcripts, (2) build an initial template combining theory-led and data-led codes, (3) apply the template to remaining data plus new inductive codes, (4) revise the template, (5) cluster codes into themes, (6) abstract themes back onto the ADI+ framework.
- **Stages 3–4 loop repeatedly (an "abductive" process) — the template is not fixed once and then mechanically applied; it keeps being reorganised as new data complicates it.**



From Quote to Framework, One Row at a Time

- The coding process is concrete, not abstract — each quote is traced through four columns to its final theoretical home:
  - Quote: "Main benefit is that we have been able to survey and include information from many small communities in the DPR." (Landscape steward, FGD 4)
  - Code: "Inclusion of demands of underrepresented groups"
  - First-order theme: "Greater representation and systematisation through participatory processes"
  - **Abstraction to ADI+ framework: "Design equity: participatory design"**
- *Parallel coding was common: a single quote often informed two different codes at once (e.g., a data-visualisation quote spoke to both design-related and institutional-shift codes) — the analysis explicitly allows this rather than forcing one code per excerpt.*

Connecting Themes Back to the Research Questions

- On completion of coding, sub-codes and codes were examined together to develop themes — an interpretive step, not a mechanical count.
- Example: the codes "people are only interested in private demands" and "commons demands are always ignored" were connected into a single theme, "Planning norms privilege private over commons resources."
- Preliminary themes were then clustered into overarching/core themes, and finally abstracted into the ADI+ framework's structural mechanisms — the causes of equity (design, resource, institutional) and the drivers of incorporation (coercion vs. choice).
- **This is the step where Step 2's operationalized sub-questions get answered: each of D1.1–D1.5 in the Findings section is one cluster of themes, traced back to one sub-question, traced back to one relationship on the Step 1 stakeholder map.**

PART 6

Findings — the Method Applied

Five findings on how equity was actively built into Commons Connect's design and rollout, plus two risks of the tool still entrenching new divides.

Maps to: paper Section D (Findings), D1–D3

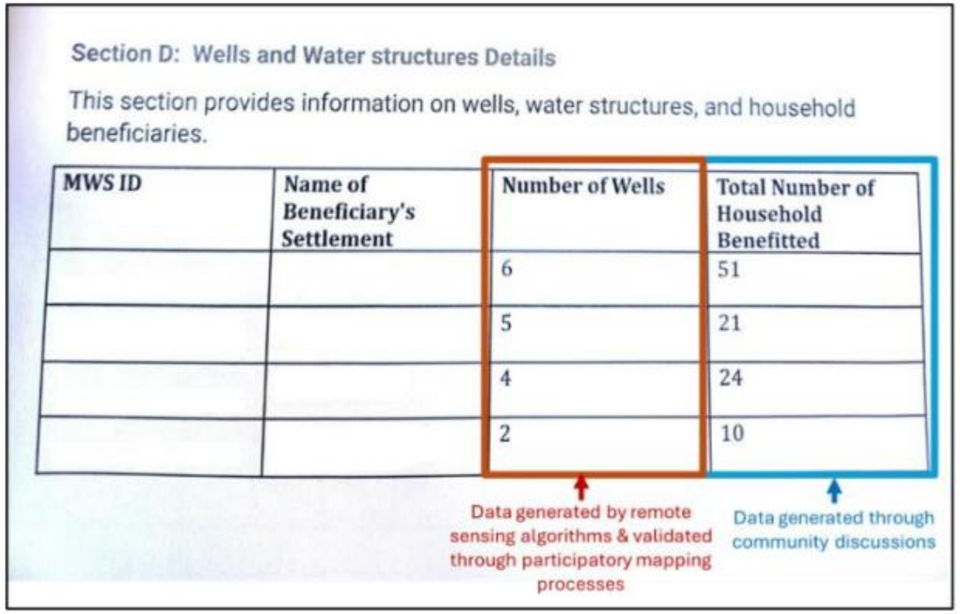
FINDING 1 (D1.1)

Decentring AI in Problem Framing

- *The design process began not with "which app to build" but with understanding MGNREGA's existing planning failures: "How is that money going to the villages within a Panchayat? ... the resource distribution of funds was very uneven... Why is that?" (CSO, Staff 2)*
- This is a "problem-discovery approach" (Smith & Lie, 2023) — understanding how the problem is shaped by social, political, and institutional dynamics before touching technology choices.
- CFPT staff explicitly framed the goal in political terms: enabling a "digital trail" of who owns which resources, as a counter to local elite capture and irregular Gram Sabha meetings.
- *"...it's about creating a digital trail of demands that have been put by the community... communities can use it as an accountability tool." (CFPT, Staff 6)*
- **Design equity, in this case, starts before any design decision — it starts with whose problem framing gets to structure the project.**

Contextualising AI: Co-producing Data with Local Knowledge

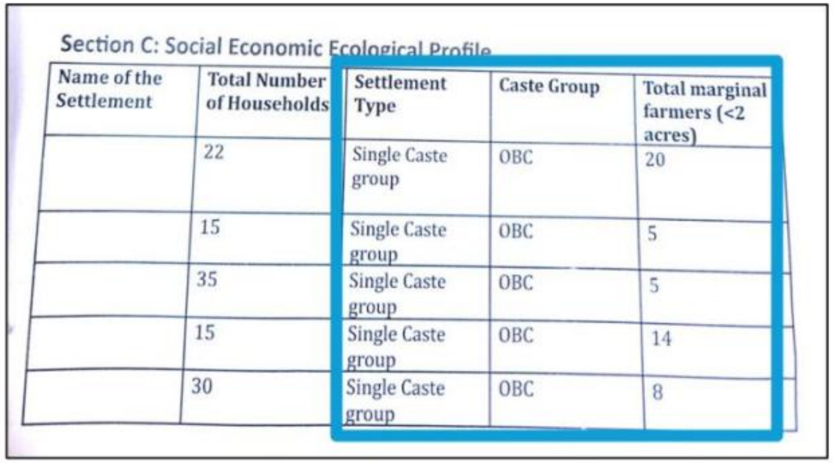
- Geospatial AI can represent physical landscapes but abstracts away the social relations through which they're accessed and contested — e.g., it can map wells, but not who owns or is excluded from them.
- Commons Connect treats AI-generated datasets as a starting point, not a complete picture: landscape stewards layer household-level data (caste, livelihoods, ownership, maintenance history) on top of remote-sensing outputs.
- *"SC has 2 wells, OBCs have 10 numbers, so how to make it equal." (Landscape steward, FGD 5) — the equity comparison only becomes visible once community-supplied caste data is layered onto the AI-detected well count.*
- **Risk named directly by the paper: AI systems that don't meaningfully involve communities in data generation and interpretation risk treating the "technical" as more real/objective than social realities like caste and power.**



Section D: Wells and Water structures Details

This section provides information on wells, water structures, and household beneficiaries.

MWS ID	Name of Beneficiary's Settlement	Number of Wells	Total Number of Household Benefitted
		6	51
		5	21
		4	24
		2	10



Section C: Social Economic Ecological Profile

Name of the Settlement	Total Number of Households	Settlement Type	Caste Group	Total marginal farmers (<2 acres)
	22	Single Caste group	OBC	20
	15	Single Caste group	OBC	5
	35	Single Caste group	OBC	5
	15	Single Caste group	OBC	14
	30	Single Caste group	OBC	8

Data generated through community discussions

Reconfiguring Legitimacy: Who Gets to Participate

- Village planning meetings were described by marginalised groups as "closed" or already-captured spaces — dominated by socially powerful actors, with SC/ST households often physically farther from the meeting point and less able to voice demands.
- *"...people from SC, ST communities, many of them are a little farther away... they don't come for meetings." (CSO Staff, FGD 5)*
- Commons Connect's field process inverted this: instead of one central Gram Sabha, stewards go settlement-to-settlement collecting demands — actively including groups who wouldn't otherwise attend a central meeting.
- *"...we are going to different settlements and conducting surveys. So, we are able to involve all groups – groups of every caste... There is an aspect of equity somewhere in this." (CSO Staff, FGD 2)*
- **In Cornwall's (2002) terms: village meetings were a "closed"/"invited-only" space; the participatory data-collection process created a genuinely "invited" space instead.**

Resource Equity: Investing in Steward Capacity

- **Access to a tool isn't the same as being able to use it meaningfully — the paper distinguishes access, usage, and outcome divides (Hammerschmidt et al., 2025).**
- Making raw data legible required deliberate translation work by CSOs, going beyond simple skills training into what the paper calls "epistemic mediation":
- *"Number wise to see the precipitation trend... that is easy. But when it's a graph, they don't understand it… so we have to refine a lot of things from our end." (CSO, Staff 3)*
- Stewards reported gaining durable new capacities — reading maps, preparing DPRs, presenting plans — work that "experts used to do", now partially democratised to a wider (if still limited) pool.
- This is resource equity in practice: not a one-time training, but sustained investment that shifts who can act on data, not just who can see it.

Non-coercion: Freedom to Distrust the Tool

- CSO staff deliberately cultivated what one described as "healthy distrust" of the tool's outputs — treating AI-generated data as provisional hypotheses requiring ground verification, not authoritative fact.
- *"...don't believe what the app tells you. If you want, go and test it..." (CSO, Staff 2)*
- *CFPT staff described conflict between the app's outputs and community knowledge as productive, not a bug: "...that conflict is okay... It will make sure that users don't completely rely on the app." (CFPT, Staff 3)*
- *Crucially, technical output was never allowed to override community judgment: "Technical reason cannot be a valid point [to reject a demand]... we have to see how reasonable their demand is." (CSO Staff, FGD 7)*
- **This is the ADI+ "drivers of incorporation" dimension in action: informed choice, not coercion — engagement with the tool's outputs is treated as optional and contestable, not binding.**

Where the Tool Could Still Entrench New Divides

- Access risk: Commons Connect launched in English, later expanded to Hindi/Gujarati/Kannada — but the key output, the Detailed Project Report (DPR), remained English-only at the time of study, limiting institutional embeddedness in local planning.
- Relational risk: the CFPT/CSO ↔ steward relationship itself carries an asymmetry — despite open-source data, stewards' capacity to use it autonomously (versus depending on CFPT/CSO interpretation) was still being built, not yet fully established.
- A further, harder limit: at the time of study it was difficult to see a path to the tool being usable by the wider community (beyond stewards) who lack the technical/domain literacy stewards have — a limitation CFPT staff acknowledged themselves.
- **The paper's framing matters here: these are read as pilot-stage growing pains worth watching closely, not as evidence the equity-oriented design failed — but the paper is explicit that even equity-oriented interventions are not automatically immune to reproducing inequality.**

PART 7

Step 5 — Discussion: Connecting to Broader Theory

The final methodological step: placing this single case's findings into conversation with the wider literature on AI and inequality.

Maps to: paper Section E (Discussion)

Beyond Algorithmic Bias: Centring Structural Inequality

- Much AI-ethics scholarship centres on the technical system itself — biased predictions, uneven access to tools (Gangadharan & Niklas, 2019) — which the paper argues under-attends to the wider socio-political inequalities AI operates within.
- **Commons Connect's approach reverses the starting point: define the structural problem first (unequal MGNREGA allocation, elite capture), name who is most affected (marginalised castes excluded from planning), and only then ask what kind of AI — if any — could help.**
- *The paper's phrase for this: you cannot lead from "what AI can do" — you must start by "grasping the problem at its root" (Kalluri, 2020).*
- This connects directly back to this course's own framing (Deck 1's Topic 1, Problem Discovery) — the paper is, in effect, independent empirical validation that problem-first design is what produced Commons Connect's equity-oriented outcomes.

Restraint as a Deliberate Counter to AI Hype

- Despite Commons Connect's datasets being technically more sophisticated than comparable tools, implementation consistently avoided treating that technical capability as the primary source of value.
- This is framed as resisting AI "hyper-visibility" — the pressure for AI to be publicly "staged" as transformative even when its practical utility is uncertain (Klingebiel, 2025; Markelius et al., 2024).
- **Restraint operates on two levels: practical (don't oversell the tool's accuracy or importance) and political (don't let the tool's outputs be treated as neutral, binding, or beyond community contestation).**
- "Suitability" recommended by an algorithm is not a value-free fact — it is a claim that itself needs to be negotiated against community knowledge and social norms, not treated as the final word.
- *This is a genuinely uncomfortable message for a course built around building better AI tools: sometimes the most equity-protective design choice is to deliberately keep the AI's role modest.*

What Would Each Method Miss?

- Deck 2's RCT asks: did the tool change what gets built, and who benefits, on average, across many GPs? It cannot tell you why a CFPT staff member chose to frame the problem the way they did, or what "healthy distrust" felt like to enact in the field.
- This paper's qualitative method asks: what practices and relationships produced (or risked undermining) equity, in rich detail, across 58 informants? It cannot tell you whether those practices actually moved population-level outcomes, or generalise with statistical confidence beyond these nine sites.
- **Neither method alone answers the full question a funder, policymaker, or the communities themselves might reasonably ask: is this working, and why?**

What This Study Could Not Fully Capture

- **Under-representation of the most marginalised community members' own voices: their perspectives are mediated primarily through CFPT/CSO staff and landscape stewards — a relatively privileged, more literate, more institutionally-connected subset of "community."**
- This is exactly the coverage pattern flagged earlier (Slide 11): 47 of 58 informants were reached only via FGD, and stewards themselves — while community members — are not interchangeable with the least-connected households the equity framework is ultimately concerned with.
- Equity as demonstrated here is procedural, not yet distributive: the paper is explicit that whether Commons Connect leads to actual re-distribution of MGNREGA resources toward marginalised communities remains an open, unresolved question.
- Data ownership, economic rights, and community control over data use — increasingly central concerns in equitable-AI debates — did not emerge as central themes here and are flagged as needing attention as the tool scales.

What This Means Beyond This One Case

- "Supply side" factors matter as much as design choices: donor funding models that reward scale (size, replication) over depth may actively undermine the sustained stewardship and capacity-building this study found to be essential.
- Team composition is a design decision: who is in the room when problem framing happens — technologists, CSO partners, community members, domain specialists — shapes whether structural inequities get foregrounded or treated as out of scope.
- **"Restraint" is offered as a genuinely practical design principle: a way of explicitly deciding, in advance, which decisions must stay in the domain of human/community judgment and never become fully automated or binding.**

The Reusable Methodology

1. Map the stakeholder ecosystem → 2. Turn the map into research questions → 3. Collect data (interviews, FGDs, observation) → 4. Code & analyse thematically → 5. Discuss in a broader context

- This pipeline generalises well beyond Commons Connect — it's a template for studying power dynamics around any deployed technology in a development context: map the ecosystem, ground your RQs in specific relationships, triangulate qualitative data across stakeholder groups, code with a disciplined hybrid method, and connect findings back to theory.
- **For your own course projects: the single most common failure mode is skipping Step 1 and Step 2 — jumping straight to interviews without first knowing which relationships you're actually trying to understand.**

Discussion Questions for Class

- The stakeholder map (Step 1) didn't include the state/MGNREGA bureaucracy as a distinct actor whose power might shift — only the village-council/Panchayat level. Should it have? What might that fifth relationship have surfaced?
- The paper finds that landscape stewards mediate almost everything — data collection, translation, capacity. Is the steward role itself a new site of power concentration the ADI+ framework, applied here, may be under-examining?
- "Healthy distrust" of the tool was actively cultivated by CSO staff. Is this a property of Commons Connect's design, or a property of these specific CSOs' institutional culture — and would it necessarily transfer to a different implementing partner?

References

Ganapathy, A., Heeks, R. & Iazzolino, G. "Materialising Equity in Community-Centric AI Tools: The Case against Technocracy." Centre for Digital Development Working Paper 117, University of Manchester, 2026.

Heeks, R. "Digital inequality beyond the digital divide: conceptualizing adverse digital incorporation in the global South." Information Technology for Development, 28(4), 2022.

Mehta, S.A. et al. "Initial observations from field testing of a digital participatory tool to improve water security in rural India." ICTD 2024, Nairobi, Kenya. (See Deck 1.)

Gioia, D.A., Corley, K.G. & Hamilton, A.L. "Seeking qualitative rigor in inductive research: notes on the Gioia methodology." Organizational Research Methods, 16(1), 2013.

King, N. "Doing template analysis." In Qualitative Organizational Research: Core Methods and Current Challenges. Sage, 2012.

Full reference list available in the source paper (27 references, Section "References").